

Cristiano Cavo

CURRICULUM VITAE • SOFTWARE ENGINEER • MANAGER • AGENTIC AI

Manager at PricewaterhouseCoopers, chartered Information Engineer (Italian Registry).

Polyglot backend-oriented: Python, Java, TypeScript, Kotlin — actively learning Go, Rust, Elixir, ADA. Focus on AI engineering — RAG, MCP, agentic systems — and extensible software architectures in production.

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GitHub [strawberry-code](#) LinkedIn [cristiano-cavo](#) Based in Gavi (AL) • Piedmont

WHAT I OFFER

- **Frontier AI engineering** — RAG, MCP, multi-agent systems and knowledge graphs (Zettelkasten, OKF) that actually run in production, not slides
- **Extensible architectures** — plugin systems, specification-driven development, automated quality gates, living documentation
- **6+ years of enterprise delivery** — banking, government, high-stakes fintech, systems handling 300k+ accesses/day
- **End-to-end solution architecture** — from design to client, including the "political" liaison between parties

PROFESSIONAL PATH

Oct 2025 — today

Milan • Italy

Manager — PricewaterhouseCoopers

Technical lead of **agentic software development** initiatives for the enterprise market. In charge of the evolution of a **frontier practice on applied generative AI**: RAG architectures, MCP integration, autonomous agent orchestration for banking, government and industrial use cases. Tech lead, team mentoring, client-facing role.

AI Agentic • LLM / RAG / MCP • Solution Architecture • Team Lead • Prompt Engineering • Blockchain • Fullstack

Feb 2022 — Oct 2025

Milan • Italy

Senior Associate — PricewaterhouseCoopers

Three years dedicated to an ambitious project at a **technology innovation consortium for the banking sector** for the **digitalisation and blockchainisation of financial instruments** aimed at banks and governmental institutions.

Architectural and applicative design of the platform **from scratch**: architecture, functional design, software design, direct client relationship, stakeholder support and "political" liaison between involved parties. In parallel, from 2024, kick-off of the dedicated focus on prompt engineering and AI agents, joining the company's GenAI practice.

Development on **Corda R3** (v4 and v5), Kotlin, Python, Cloud SaaS.

Banking R&D Consortium • Solution Architect • Stakeholder Mgmt • Finance / Government • Corda 5 • Kotlin • Python • Cloud SaaS • Prompt Engineer

Dec 2019 — Feb 2022

Milan • Italy

Associate — PricewaterhouseCoopers

Two years dedicated to one of the **largest Italian and European banks** for employee access management: web platform, server, mobile, DevOps. I came to oversee the **entire end-to-end platform** of the service, which today stably handles **over 300,000 daily accesses** without interruption. In parallel, blockchain timestamping solutions for fashion, automotive and nautical clients.

Banking • Tier 1 EU • Identity & Access Mgmt • 300k+ accessi/giorno • React Native • JS / TS • Python • DevOps • Mobile (iOS/Android/Huawei/WP) • Blockchain Timestamping • Cybersecurity

Oct 2019 — Nov 2019

Turin • Italy

Researcher — LINK Foundation

Research activity on **Hyperledger Suite** and Elastic Search indexing for enterprise blockchain solutions.

Research • Blockchain • Hyperledger • Elastic Search

Aug 2019 — Sep 2019

Turin · Italy

Software Developer — InteK AI

Development of a **C# DLL library** to integrate the **NEO** blockchain (formerly Antshares) with the electronic electoral systems of Cyprus. Hands-on exposure to the **.NET ecosystem** and Visual Studio in a governmental context.

C# / .NET · Visual Studio · NEO Blockchain · Government · IoT

Apr 2018 — Jun 2019

Turin · Italy

Research Grant — Politecnico di Torino

Research on **databases and storage technologies for blockchain**, IPFS, distributed systems. Support for the publication of a scientific paper on distributed database types.

Research · Databases · Blockchain · IPFS

Sep 2018 — Apr 2019

Turin · Italy

M.Sc. Thesis — Politecnico di Torino · General Motors

Supporting **Connected Manufacturing using Blockchain in the Automotive Domain**. Architecture design, development of a working prototype, integration of three **Industry 4.0** enabling technologies: IoT, 3D printing and blockchain. Solidity smart contracts on Ethereum, MakerBot Replicator interfaced with a Raspberry Pi.

Automotive · Industria 4.0 · Ethereum / Solidity · 3D Print · IoT · Software Architecture · Traceability

Apr 2018 — Sep 2018

Turin · Italy

Software Developer — CSP

Health and environmental sustainability sectors. PHP fullstack development (SLIM micro-framework), gamification with Unity 3D for hospital rehabilitation projects (T.R.E.E.), IoT with MQTT.

PHP / SLIM · Unity 3D · C# / WPF · Health · IoT / MQTT · Gamification

FOCUS 2024–2026

The company I'm currently working at is investing significant resources in this vertical and our team is **frontier** on the topic in Italy. We apply it **every day**, keeping ourselves updated on the weekly evolution of the ecosystem (Anthropic, OpenAI, MCP standard, orchestration frameworks).

01 / ARCHITECTURES

RAG on enterprise corpora — Ingestion pipelines, chunking, hybrid retrieval (semantic + lexical), grounding, citation tracking.

02 / PROTOCOLS

MCP server & client — Exposing corporate tools and resources to AI agents via Model Context Protocol.

03 / ORCHESTRATION

Autonomous agents — Planning, execution and evaluation loops. Multi-step workflows with human-in-the-loop.

04 / KNOWLEDGE

Knowledge Graph & Zettelkasten — Graph engineering and the Zettelkasten method for OKF knowledge bases, agent-queryable.

STACK & ENGINEERING

LLM & AI ENGINEERING

RAG Architecture — Vector search, hybrid search, graph RAG

Agentic Systems — Tool use, multi-agent orchestration, planning loops

Model Context Protocol — Server, client, integrations in production

Multi-LLM Abstraction — LiteLLM, Rig-core, model routing

Embedding Pipelines — Semantic indexing, chunking, citation tracking

BACKEND · API · DATA

API Design REST/OpenAPI-first — Automatic client and schema generation

Backend frameworks — FastAPI, Axum (Tokio), Express

Async Task Processing — Celery, Tokio, background queues, event-driven microservices

Relational persistence — PostgreSQL, SQLAlchemy, query optimisation, SQLite (edge)

Specialised persistence — Vector DB (Qdrant), Graph DB (Neo4j), full-text (Elasticsearch), Redis

FRONTEND · DEVOPS · TESTING

Frontend Stack — React/Next.js, TypeScript strict, Tailwind, Shadcn

Modern tooling — Bun runtime, monorepo (Bun + Cargo workspaces), uv, Cargo, Go modules

Container & Deploy — Docker/Compose, Vercel, Raspberry Pi edge deployment

Quality Gates — MyPy, TypeScript strict, Ruff, Clippy, ESLint, pre-commit hooks

Testing — Pytest + pytest-asyncio, Playwright E2E, Cargo test, pragmatic TDD

EDUCATION · REGISTRY · LANGUAGES

EDUCATION

M.Sc. in Computer Engineering — Software specialist · Politecnico di Torino April 2019

B.Sc. in Computer Engineering — Politecnico di Torino October 2015

High School Diploma — Scientific Lyceum — Liceo Edoardo Amaldi · Novi Ligure July 2010

QUALIFICATIONS

Chartered Information Engineer — Registered with the Order of Engineers of Alessandria · Section A April 2023

Prof. Fabrizio Lamberti · Politecnico di Torino

Prof. Valentina Gatteschi · Politecnico di Torino

LANGUAGES

Italian — Native

English — Professional · B2/C1

French — Elementary · A2

INSTITUTIONAL ACTIVITY & CULTURAL INTERESTS

Founding member — Aerospace Commission

Order of Engineers of the Province of Alessandria

I have been participating for about a year in an institutional initiative aimed at stimulating the **conversion of the Alessandria industrial fabric** from the automotive sector to aerospace and Defense, in response to the automotive crisis and the current geopolitical instability.

Geopolitics, strategic studies and dual-use technology

Systematic reading · personal interest

For several years I have devoted constant attention to **geopolitical scenarios**, international security dynamics, Defense industrial doctrines, technological competition between blocs and **dual-use policies**.